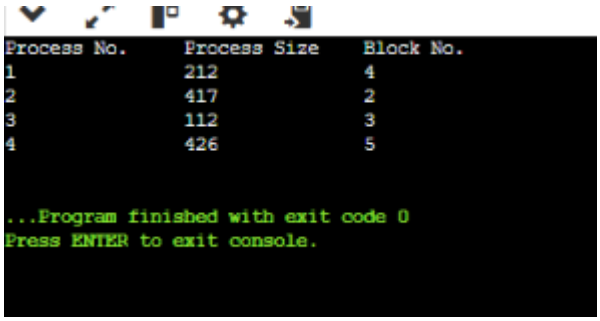
EXPERIMENT 22: Best Fit Memory Allocation

code:

```
#include <stdio.h>

int main() {

  int blockSize[] = {100, 500, 200, 300, 600};
  int processSize[] = {212, 417, 112, 426};

  int allocation[4];

  int m = 5, n = 4;

  for (int i = 0; i < n; i++)
    allocation[i] = -1;

  for (int i = 0; i < n; i++) {

    int best = -1;

    for (int j = 0; j < m; j++) {

      if (blockSize[j] >= processSize[i]) {

        if (best == -1 ||
          blockSize[j] < blockSize[best]) {

          best = j;
        }
      }
```

```
        }

        if (best != -1) {

            allocation[i] = best;

            blockSize[best] -= processSize[i];
        }
    }

    printf("Process No.\tProcess Size\tBlock No.\n");

    for (int i = 0; i < n; i++) {

        printf("%d\t\t%d\t\t", i + 1, processSize[i]);

        if (allocation[i] != -1)
            printf("%d\n", allocation[i] + 1);
        else
            printf("Not Allocated\n");
    }

    return 0;
}
```

Output: